

Unit V: Fourier Transforms

Class 8:

1. Find If $\mathcal{F}\{f(t)\} = \sqrt{\pi}e^{\frac{-\omega^2}{4}}$ for $f(t) = e^{-t^2}$, find Fourier transform of

a) $e^{\frac{-x^2}{a}}$

b) $e^{-4(x-a)^2}$

2. If $\mathcal{F}[e^{-5t}u(t-3)] = \frac{e^{-b(a+i\omega)}}{3+i\omega}$ find the value of $a - b$.
